Tensor decompositions for data analysis

Module 2 of 4

Applied Math 210: Algebraic Fundamentals of Representing Data

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Introduction

The first part of the course studied factorizations of a matrix, such as the covariance matrix or data matrix. We saw that factorizations of the covariance matrix arise when modeling an observed random vector $\mathbf{x} \in \mathbb{R}^p$ as a linear transformation of a latent random vector $\mathbf{z} \in \mathbb{R}^q$, via $\mathbf{x} = A\mathbf{z}$. We recovered A from the covariance $\Sigma_{\mathbf{x}} = A\Sigma_{\mathbf{z}}A^{\mathsf{T}}$ under the assumption that the entries of $\mathbf{z}$ are *uncorrelated*. This module of the course replaces "uncorrelated" with something stronger, the assumption that the entries of $\mathbf{z}$ are independent. We study independence by looking beyond the covariance matrix, at higher-order statistics. The higher-order statistics are *tensors*: multi-indexed arrays that generalize matrices. A random vector's covariance is its second cumulant; its higher-order cumulants (third, fourth, etc.) are higher-order tensors. Diagonalizing these tensor underpins many data analysis methods. Tensors also arise as arrays of data that are organized via multiple indices, such as a tensor of neurons by experiment by time in neuroscience.

This module studies the multi-linear algebra of tensors: what they are, what their rank means, and how existence and uniqueness of a tensor decomposition are proved. It then studies independent component analysis (ICA), which is the factorization $\mathbf{x} = A\mathbf{z}$ where the entries of $\mathbf{z}$ are mutually independent. We prove an identifiability theorem due to Comon: an ICA model is identifiable if and only if at most one source is Gaussian. The same three ideas from Module 1 (expressivity, identifiability, optimization landscapes) reappear, with an improvement to identifiability: for tensors of order at least three, a generic low-rank decomposition is unique, with no extra structural assumption (such as orthogonality) required.

The material in this part of the course lays the foundation for tensor decomposition algorithms (such as alternating least squares, Jennrich's algorithm, or tensor power iteration), and the use of moment and cumulant methods to estimate latent-variable and mixture models beyond ICA.

Useful texts

This module studies the multi-linear algebra of tensors, motivated by identifying a linear latent-variable model from higher-order statistics or by finding factors in tensors of data. J. M. Landsberg's *Tensors: Geometry and Applications* studies tensor rank, symmetric rank, and the algebraic geometry of secant varieties. Kolda and Bader's survey *Tensor Decompositions and Applications* and Hyvärinen, Karhunen, and Oja's book *Independent Component Analysis* state the results we prove here (regarding uniqueness of tensor decomposition and of ICA). Lek-Heng Lim's *Tensors in computations* (Acta Numerica, 2021) gives the definition of a tensor via three ideas (equivariance, multilinearity, separability) a useful perspective on our starting point, which will be the definitions of a tensor. A related reference is my article "Factorizations for Data Analysis," *SIAM News*, Vol. 58, No. 7, 2025, which covers ICA, briefly.

Contents

1 Notation

We continue Module 1's notation conventions: bold lowercase for vectors (fixed or random), plain lowercase for scalars, plain uppercase for matrices. Tensors of any order are also set in plain uppercase. A subscript on a bold symbol labels a vector, and a superscript in parentheses labels which factor of a tensor (or vector space) is meant.

Symbol	Meaning
$\mathbf{x} = (x_1, \ldots, x_p)^\mathsf{T}$	random vector of p observed variables
$\mathbf{z} = (z_1, \ldots, z_q)^\mathsf{T}$	random vector of q latent (source) variables
$A \in \mathbb{R}^{p\times q}$	mixing matrix, $\mathbf{x} = A\mathbf{z}$; column $\mathbf{a}_k$ is source k's loading onto each variable
$T \in \mathbb{R}^{n_1} \otimes \cdots \otimes \mathbb{R}^{n_d}$	a tensor of order d and format $n_1 \times \cdots \times n_d$; entries $T_{i_1\ldots i_d}$
$S^d(\mathbb{R}^n)$	the space of symmetric order-d tensors on $\mathbb{R}^n$
$\mathbf{u}^{(1)} \otimes \cdots \otimes \mathbf{u}^{(d)}$	a rank-one tensor; entries $u^{(1)}_{i_1} \cdots u^{(d)}_{i_d}$
$\mathbf{v}^{\otimes d}$	the symmetric rank-one tensor $\mathbf{v} \otimes \cdots \otimes \mathbf{v}$ (d times)
$\operatorname{rank}(T)$	tensor rank: fewest rank-one summands
$\operatorname{rank}_+(T)$	non-negative rank: fewest rank-one summands with non-negative factors
k_A	Kruskal rank of A: largest k such that every k columns of A are linearly independent
$\mathcal{M}_d(\mathbf{z}) \in S^d(\mathbb{R}^q)$	d-th moment tensor of $\mathbf{z}$, $\mathbb{E}[\mathbf{z}^{\otimes d}]$
$\mathcal{K}_d(\mathbf{z}) \in S^d(\mathbb{R}^q)$	d-th cumulant tensor of $\mathbf{z}$, $\mathbb{E}[\mathbf{z}^{\otimes d}]$ up to lower order terms
$A \bullet S$	the action of $A \in \mathbb{R}^{p\times q}$ on $S \in S^d(\mathbb{R}^q)$, generalizing the congruence $A\Sigma A^\mathsf{T}$
$T(\mathbf{u}, \mathbf{v}, \mathbf{w})$	multilinear-map evaluation, $\sum_{ijk} T_{ijk} u_i v_j w_k$ (similarly for other orders)

2 What Is a Tensor?

To define a tensor, it helps to first reflect on how best to define a matrix, and then to take its order-d generalization. A matrix $M \in \mathbb{R}^{n_1 \times n_2}$ is:

(i) a 2D array of numbers, $(M_{ij})_{1 \le i \le n_1, 1 \le j \le n_2}$;

(ii) a linear map $\mathbb{R}^{n_2} \to \mathbb{R}^{n_1}$;

(iii) an element of the tensor product space $\mathbb{R}^{n_1} \otimes \mathbb{R}^{n_2}$, built from the outer product $\mathbf{u} \otimes \mathbf{v} := \mathbf{u}\mathbf{v}^\mathsf{T}$ of two vectors $\mathbf{u} \in \mathbb{R}^{n_1}$, $\mathbf{v} \in \mathbb{R}^{n_2}$, via $(\mathbf{u} \otimes \mathbf{v})_{ij} = u_i v_j$.

A tensor of order d is the same object with d indices instead of two: a d-way array; a multilinear map of $d-1$ vector arguments; an element of a d-fold tensor product space.

Definition 2.1 (Tensor, order, format)**.** A *tensor of order* d is an array T with entries $T_{i_1 \dots i_d}$, $i_1 \in \{1, \dots, n_1\}, \dots, i_d \in \{1, \dots, n_d\}$. We say T has *format* $n_1 \times \cdots \times n_d$, and write $T \in \mathbb{R}^{n_1} \otimes \cdots \otimes \mathbb{R}^{n_d}$.

Order $d = 1$ is a vector; order $d = 2$ is a matrix.

Definition 2.2 (Outer product, tensor product space)**.** For vectors $\mathbf{u}^{(1)} \in \mathbb{R}^{n_1}, \dots, \mathbf{u}^{(d)} \in \mathbb{R}^{n_d}$, their *outer product* $\mathbf{u}^{(1)} \otimes \cdots \otimes \mathbf{u}^{(d)}$ is the order-d tensor with entries

$$\left(\mathbf{u}^{(1)} \otimes \cdots \otimes \mathbf{u}^{(d)}\right)_{i_1 \dots i_d} = u^{(1)}_{i_1} \cdots u^{(d)}_{i_d}.$$

The outer product is multilinear: $(\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2) \otimes \mathbf{v} = \lambda_1(\mathbf{u}_1 \otimes \mathbf{v}) + \lambda_2(\mathbf{u}_2 \otimes \mathbf{v})$, and likewise in the other argument. The *tensor product space* $\mathbb{R}^{n_1} \otimes \cdots \otimes \mathbb{R}^{n_d}$ is the space of all finite sums $\sum_k \mathbf{u}^{(1)}_k \otimes \cdots \otimes \mathbf{u}^{(d)}_k$.

A tensor is also a multilinear map: T of format $n_1 \times \cdots \times n_d$ defines $T(\mathbf{u}^{(1)}, \dots, \mathbf{u}^{(d)}) := \sum_{i_1 \dots i_d} T_{i_1 \dots i_d} u^{(1)}_{i_1} \cdots u^{(d)}_{i_d} = \langle T, \mathbf{u}^{(1)} \otimes \cdots \otimes \mathbf{u}^{(d)} \rangle$, which is linear in each argument (where here $\langle A, B \rangle = \sum A_{i_1 \dots i_d} B_{i_1 \dots i_d}$ is the entrywise inner product). Equivalently, T can be viewed as a linear map $\mathbb{R}^{n_1} \otimes \cdots \otimes \mathbb{R}^{n_{d-1}} \to \mathbb{R}^{n_d}$. The matrix case ($d = 2$) is the special case with one index in the domain. For a tensor, there are more choices of domain and codomain: a tensor is equivalently viewed as a multilinear map in which some subset of indices appear in the domain and the complement set in the codomain.

Definition 2.3 (Tensor rank)**.** A non-zero tensor T has *rank one* if $T = \mathbf{u}^{(1)} \otimes \cdots \otimes \mathbf{u}^{(d)}$ for some vectors $\mathbf{u}^{(i)}$. It has *rank* r if $T = \sum_{k=1}^{r} \mathbf{u}^{(1)}_k \otimes \cdots \otimes \mathbf{u}^{(d)}_k$, and r is minimal such that such a decomposition exists.

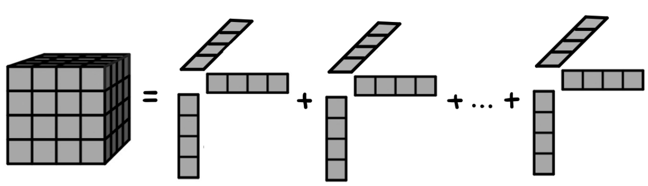

Figure 1: A tensor written as a sum of outer products of vectors.

Definition 2.4 (Symmetric tensor, symmetric rank)**.** A tensor $T \in (\mathbb{R}^n)^{\otimes d}$ (i.e. $n_1 = \cdots = n_d = n$) is *symmetric* if $T_{i_1 \ldots i_d} = T_{\sigma(i_1)\ldots\sigma(i_d)}$ for every permutation σ of $\{1, \ldots, d\}$. We write $S^d(\mathbb{R}^n)$ for the space of such tensors. A rank-one tensor is symmetric iff it has the form $\mathbf{v}^{\otimes d} := \mathbf{v} \otimes \cdots \otimes \mathbf{v}$ for some $\mathbf{v} \in \mathbb{R}^n$. A symmetric tensor S has *symmetric rank* r if $S = \sum_{k=1}^{r} \lambda_k \mathbf{v}_k^{\otimes d}$ for scalars λ_k and unit vectors $\mathbf{v}_k$, and r is fewest such.

Exercise 2.5. Check that the above definitions specialize to the definition of matrix rank.

For symmetric matrices, symmetric rank equals ordinary matrix rank (see Section 4, Remark 4.5). This is known to fail for real tensors, but remains open over the complex numbers.

3 Examples of Tensors

Tensor rank can have various interpretations. We consider four examples.

Data Tensors

While a typical dataset is organized as samples by variables, many datasets are naturally indexed by three or more categories. These include: fluorescence spectroscopy (samples $\times$ emission wavelengths $\times$ excitation wavelengths), color images (height $\times$ width $\times$ RGB channel), or neural recordings (neuron $\times$ experiment $\times$ time). As with a data *matrix*, where rank counts the number of factors combined in the data, the rank of a data *tensor* counts the number of underlying signals combined across all indexed contexts at once.

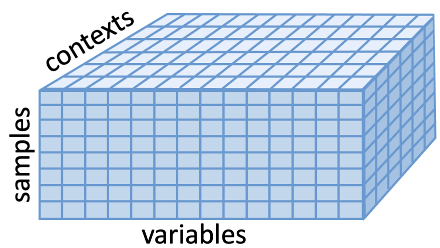

Figure 2: A data tensor, samples as rows, variables as columns, tubes as contexts

Polynomials

Proposition 3.1 (Correspondence between symmetric tensors and homogeneous polynomials)**.** *There is a linear bijection between $S^d(\mathbb{C}^n)$ and homogeneous degree-d polynomials in n variables, sending $S \mapsto f_S$,*

$$f_S(\mathbf{x}) = \sum_{i_1,\ldots,i_d} S_{i_1\ldots i_d} x_{i_1} \cdots x_{i_d}, \qquad \mathbf{x} = (x_1, \ldots, x_n)^{\mathsf{T}}.$$

We work over $\mathbb{C}$ in this section, since the genericity statements below (Alexander–Hirschowitz, Section 4) are true there. The map is well defined and linear by construction; it is a bijection because every homogeneous degree-d polynomial has a unique expansion in the monomial basis, and symmetry of S is the requirement needed to recover the entries of S from those coefficients (they are equal up to multinomial coefficients). If T is not symmetric, $T \mapsto f_T$ is still well defined but no longer injective: any tensor with the same symmetrization gives the same polynomial.

The polynomial f corresponding to the rank one tensor $\lambda\mathbf{v}^{\otimes d}$ is $\lambda\langle\mathbf{v},\mathbf{x}\rangle^d$, a d-th power of a linear form. Extending by linearity:

Corollary 3.2. *A symmetric tensor $S \in S^d(\mathbb{C}^n)$ has symmetric rank $\leq r$ if and only if f_S is a sum of r d-th powers of linear forms (a* Waring decomposition *of f_S).*

A homogeneous polynomial of degree d in $n+1$ variables $x_1,\ldots,x_n,x_{n+1}$ defines, via $x_{n+1}=1$, a polynomial of degree $\leq d$ in n variables; its zero set is a hypersurface in $\mathbb{R}^n$ (or $\mathbb{C}^n$). A degree-3 hypersurface is a *cubic surface*.

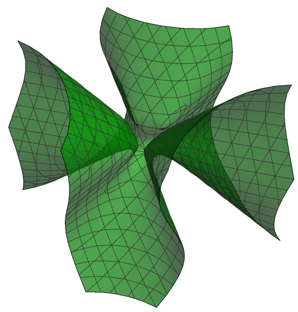

Figure 3: An example of a cubic surface

Theorem 3.3 (Sylvester's Pentahedral Theorem, 1851)**.** *A general cubic surface f (equivalently, a general $T \in S^3(\mathbb{C}^4)$) can be written as*

$$f = \ell_1^3 + \cdots + \ell_5^3$$

for linear forms $\ell_1,\ldots,\ell_5$ with complex coefficients.

This is consistent with the generic-rank formula we'll get to later in Section 4: $\dim S^3(\mathbb{C}^4) = \binom{6}{3} = 20$, and $\lceil 20/4 \rceil = 5$.

Probability Mass Functions

Let $X_1,\ldots,X_d$ be discrete random variables, with X_k taking m_k values, and let P be the joint probability mass function, viewed as a tensor of format $m_1\times\cdots\times m_d$: $P_{i_1\ldots i_d} = \Pr(X_1=i_1,\ldots,X_d=i_d)$.

Proposition 3.4. *P has rank 1 if and only if $X_1,\ldots,X_d$ are independent.*

Proof. We give the argument for $d=2$ for notational ease. If X_1,X_2 independent, $P_{ij} = \Pr(X_1 = i)\Pr(X_2=j) = \mathbf{u}\mathbf{v}^\mathsf{T}$ for $u_i = \Pr(X_1=i)$, $v_j = \Pr(X_2=j)$: rank 1.

Conversely, suppose $P = \mathbf{u}\mathbf{v}^\mathsf{T}$ for some (not necessarily non-negative) $\mathbf{u},\mathbf{v}$. Since $P_{ij} = u_iv_j \geq 0$ for all i,j, fixing any i_0 with $u_{i_0}\neq 0$ forces every v_j to share the sign of $u_{i_0}^{-1}$; then every u_i must in turn share a matching sign for $u_iv_j\geq 0$ to hold for all j. So $u_iv_j = |u_i|\,|v_j|$ for all i,j: $P = \mathbf{u}'(\mathbf{v}')^\mathsf{T}$ with $\mathbf{u}' = |\mathbf{u}|, \mathbf{v}' = |\mathbf{v}|$ entrywise non-negative. Since $\sum_{ij} P_{ij} = 1$, $\|\mathbf{u}'\|_1\|\mathbf{v}'\|_1 = 1$. Setting $\mathbf{u}'' = \mathbf{u}'/\|\mathbf{u}'\|_1$, $\mathbf{v}'' = \mathbf{v}'/\|\mathbf{v}'\|_1$ gives $\mathbf{u}''(\mathbf{v}'')^\mathsf{T} = \mathbf{u}'(\mathbf{v}')^\mathsf{T}/(\|\mathbf{u}'\|_1\|\mathbf{v}'\|_1) = P$, and $u_i'' = \sum_j P_{ij} = \Pr(X_1=i)$, $v_j'' = \Pr(X_2=j)$: $P_{ij} = \Pr(X_1=i)\Pr(X_2=j)$, i.e. independence. ☐

Non-negativity of the factors is required to extend the above to higher ranks.

Definition 3.5 (Non-negative rank)**.** T has *non-negative rank* 1 if $T = \mathbf{u}^{(1)}\otimes\cdots\otimes\mathbf{u}^{(d)}$ with every $\mathbf{u}^{(i)}$ entrywise non-negative. T has *non-negative rank* r if it is a sum of r non-negative-rank-1 tensors, and not fewer.

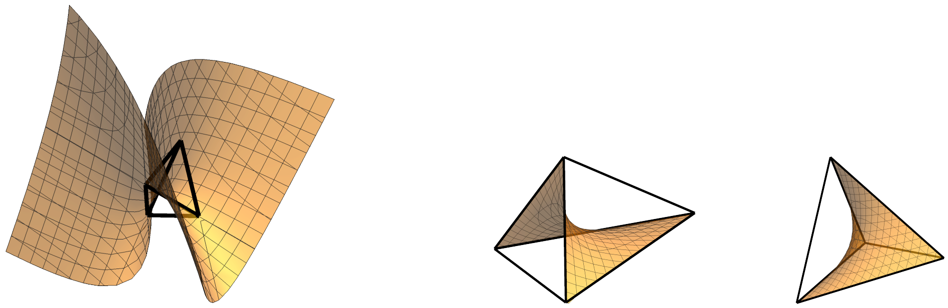

Figure 4: Left: the set of rank one 2×2 tensors, showing their intersection with the set of tensors with non-negative entries that sum to 1. Middle and right: two views of the independence model: the set of rank one non-negative tensors that sum to 1.

Proposition 3.6. *If $X_1, \dots, X_d$ are conditionally independent given an unobserved variable Z with r states, then P has non-negative rank $\leq r$. Consequently, for two variables, P has non-negative rank $\leq r$ if and only if X_1, X_2 are conditionally independent given some r-state Z.*

Proof. For $d = 2$: writing $\lambda_k = \Pr(Z = k)$, $a_{ik} = \Pr(X_1 = i \mid Z = k)$, $b_{jk} = \Pr(X_2 = j \mid Z = k)$,

$$P_{ij} = \sum_{k=1}^{r} \Pr(X_1 = i, X_2 = j \mid Z = k) \Pr(Z = k) = \sum_{k=1}^{r} \lambda_k \, a_{ik} \, b_{jk} = \Big(\sum_{k=1}^{r} \lambda_k \, \mathbf{a}_k \otimes \mathbf{b}_k \Big)_{ij},$$

a sum of r non-negative rank-one terms. Conversely, given $P = \sum_{k=1}^{r} \lambda_k \mathbf{a}_k \otimes \mathbf{b}_k$ with $\lambda_k \geq 0$, $\sum_k \lambda_k = 1$, and each $\mathbf{a}_k, \mathbf{b}_k$ a probability vector, introduce a latent Z with $\Pr(Z = k) = \lambda_k$, $\Pr(X_1 = i \mid Z = k) = a_{ik}$, $\Pr(X_2 = j \mid Z = k) = b_{jk}$. This yields conditional independence. $\square$

Exercise 3.7 (Rank and non-negative rank differ)**.** Let $M = \begin{pmatrix} 1&1&0&0 \\ 1&0&1&0 \\ 0&1&0&1 \\ 0&0&1&1 \end{pmatrix}$. Show that $\operatorname{rank}(M) = 3$ and that $\operatorname{rank}_+(M) = 4$.

Matrix Multiplication

Definition 3.8 (Matrix multiplication tensor)**.** For $A \in \mathbb{R}^{n_1 \times n_2}$, $B \in \mathbb{R}^{n_2 \times n_3}$, $C \in \mathbb{R}^{n_3 \times n_1}$, define the trilinear form

$$T_{\langle n_1, n_2, n_3 \rangle}(A, B, C) := \operatorname{tr}(ABC).$$

Viewed as an element of $(\mathbb{R}^{n_1 \times n_2})^* \otimes (\mathbb{R}^{n_2 \times n_3})^* \otimes (\mathbb{R}^{n_3 \times n_1})^*$, this is the *matrix multiplication tensor*, of format $n_1 n_2 \times n_2 n_3 \times n_1 n_3$.

Proposition 3.9. *We have the upper bound* $\operatorname{rank}\big(T_{\langle n_1, n_2, n_3 \rangle}\big) \leq r$ *if and only if there is an algorithm computing the entries of AB using r products of the form $\alpha_k(A)\beta_k(B)$, for linear forms α_k, β_k, followed only by additions.*

Proof sketch. Suppose $T_{\langle n_1, n_2, n_3 \rangle} = \sum_{k=1}^{r} \alpha_k \otimes \beta_k \otimes \gamma_k$, so $\operatorname{tr}(ABC) = \sum_{k=1}^{r} \alpha_k(A)\beta_k(B)\gamma_k(C)$ for all A, B, C. Fix A, B and set $\mu_k = \alpha_k(A)\beta_k(B)$ (computed with r products); then $\operatorname{tr}(ABC) = \sum_k \mu_k \gamma_k(C)$ for every C. Taking C to be the elementary matrix E_{li} gives $\operatorname{tr}(ABE_{li}) = (AB)_{il}$, so $(AB)_{il} = \sum_k \mu_k \, \gamma_k(E_{li})$: every entry of AB is a fixed linear combination of the r products μ_k. Conversely, an algorithm of this form determines such a rank-r decomposition. $\square$

The naive algorithm uses $r = n_1 n_2 n_3$ products (one per entry of A times one per entry of B, summed). For 2×2 matrices ($n_1 = n_2 = n_3 = 2$), this is $r = 8$.

Theorem 3.10 (Strassen, 1969)**.** $\operatorname{rank}\big(T_{\langle 2,2,2\rangle}\big) \leq 7$.

Proof. Exhibit the 7 products directly: with $A = \left(\begin{smallmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{smallmatrix}\right)$, $B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$, set

$$\begin{aligned} &M_1 = (a_{11}+a_{22})(b_{11}+b_{22}), && M_2 = (a_{21}+a_{22})b_{11}, && M_3 = a_{11}(b_{12}-b_{22}), \\ &M_4 = a_{22}(b_{21}-b_{11}), && M_5 = (a_{11}+a_{12})b_{22}, && M_6 = (a_{21}-a_{11})(b_{11}+b_{12}), \\ &M_7 = (a_{12}-a_{22})(b_{21}+b_{22}). \end{aligned}$$

One checks that $(AB)_{11} = M_1 + M_4 - M_5 + M_7$, $(AB)_{12} = M_3 + M_5$, $(AB)_{21} = M_2 + M_4$, $(AB)_{22} = M_1 - M_2 + M_3 + M_6$; by Proposition 3.9, this is a rank-7 decomposition of $T_{\langle 2,2,2\rangle}$. □

It is an open problem to determine $\operatorname{rank}(T_{\langle n_1,n_2,n_3\rangle})$ in general. Recent work has used AI to tackle this problem for fixed n_i. In 2022, AlphaTensor (Fawzi et al., DeepMind) used reinforcement learning to find a new low-rank decompositions of $T_{\langle n_1,n_2,n_3\rangle}$ for several small formats. In 2025, AlphaEvolve (DeepMind) found a rank-48 decomposition of $T_{\langle 4,4,4\rangle}$ for complex matrix multiplication, the first improvement since Strassen (1969) on the rank-49 bound obtained by recursively applying the $T_{\langle 2,2,2\rangle}$ decomposition.

Definition 3.11. The *matrix multiplication exponent* ω is the infimum c such that $n \times n$ matrices can be multiplied using $O(n^c)$ arithmetic operations.

Recursively applying a rank-r decomposition of $T_{\langle n,n,n\rangle}$ in blocks gives an algorithm for $N \times N$ matrix multiplication using $O(N^{\log_n r})$ operations; this connects Proposition 3.9 to ω. The naive algorithm gives $\omega \leq 3$. Strassen's rank-7 decomposition of $T_{\langle 2,2,2\rangle}$ gives $\omega \leq \log_2 7 \approx 2.807$. Refinements, including Coppersmith and Winograd (1990, $\omega < 2.376$), and a sequence of "laser method" improvements since, have brought the best known bound down to $\omega < 2.371339$ (Alman, Duan, Vassilevska Williams, Xu, Xu, and Zhou, 2025). Whether $\omega = 2$ (whether matrix multiplication is asymptotically as easy as reading the input) is an open problem.

4 The Generic Rank of a Tensor

Definition 4.1 (Genericity)**.** A property of points in $\mathbb{C}^N$ holds *generically* if it holds away from a lower-dimensional (Zariski-closed) set. This implies that it holds with probability 1 under any continuous distribution on parameters.

For example: a generic $x \in \mathbb{R}$ is non-zero; for a fixed non-zero polynomial $f(x_1, \ldots, x_n)$, a generic point $\mathbf{p} \in \mathbb{R}^n$ has $f(\mathbf{p}) \neq 0$ (the zero set of a non-zero polynomial is always lower-dimensional). We ask: what is the rank of a generic tensor?

Warm-Up: Symmetric Matrices

We first study this for $d = 2$, by counting parameters in the eigendecomposition. Consider $S = \sum_{i=1}^r \lambda_i \mathbf{v}_i \mathbf{v}_i^\mathsf{T} \in S^2(\mathbb{R}^p)$ with the $\mathbf{v}_i$ orthonormal (an eigendecomposition truncated to r terms).

Lemma 4.2. *The set of $S \in S^2(\mathbb{R}^p)$ with* $\operatorname{rank}(S) \leq r$ *has dimension $rp - \binom{r}{2}$, for $0 \leq r \leq p$.*

Proof. Parametrize by $(\lambda_1, \ldots, \lambda_r, \mathbf{v}_1, \ldots, \mathbf{v}_r)$ with $\mathbf{v}_1, \ldots, \mathbf{v}_r$ orthonormal. Choosing $\mathbf{v}_1$ (a unit vector, up to sign) costs $p-1$ parameters; choosing $\mathbf{v}_2$ orthogonal to $\mathbf{v}_1$ (a unit vector in a $(p-1)$-dimensional space) costs $p-2$; in general $\mathbf{v}_k$ costs $p-k$. Together with r parameters for $\lambda_1, \ldots, \lambda_r$, the total is

$$r + \sum_{k=1}^{r} (p-k) = r + rp - \binom{r+1}{2} = rp - \binom{r}{2}. \qquad \square$$

At $r = p$: $\dim = p^2 - \binom{p}{2} = \binom{p+1}{2} = \dim(S^2(\mathbb{R}^p))$. The rank-$\leq p$ locus is the whole space. At $r = p-1$:

$$\dim = (p-1)p - \binom{p-1}{2} = \binom{p+1}{2} - 1,$$

one less than $\dim(S^2(\mathbb{R}^p))$: the rank-$\leq p-1$ locus is a hypersurface (indeed, it is $\{S : \det S = 0\}$, the zero set of one polynomial). Hence, since we know a rank p decomposition exists from the eigendecomposition,

Corollary 4.3. *A generic $S \in S^2(\mathbb{R}^p)$ has rank p.*

An alternate argument is to use the generic uniqueness of the eigendecomposition: a generic symmetric matrix cannot have an eigendecomposition in which an eigenvalue vanishes.

Symmetric Tensors of General Order

For $d \geq 3$ we redo the parameter count, this time for the unconstrained (not necessarily orthogonal) symmetric CP decomposition $T = \sum_{i=1}^r \lambda_i \mathbf{v}_i^{\otimes d}$ used to define symmetric rank.

Lemma 4.4. *We have* $\dim S^d(\mathbb{C}^n) = \binom{n+d-1}{d}$.

Proof. A symmetric tensor's entries are determined by, and freely chosen for, each multiset of d indices from $\{1, \ldots, n\}$ (since permuting indices does not change the entry). Order such a multiset monotonically, $1 \leq i_1 \leq \cdots \leq i_d \leq n$. Represent it as a string of d marks (one per index) and $n-1$ dividers, placed among $d+n-1$ positions in total, where the divider before position k separates values $< k$ from values $\geq k$: this is a bijection between monotone index strings and ways to choose which d of the $d+n-1$ positions are marks, giving $\binom{n+d-1}{d}$ monotone strings. □

Writing λ_i's scale into $\mathbf{v}_i$, a symmetric CP expression $T = \sum_{i=1}^r \mathbf{v}_i^{\otimes d}$ has rn free parameters (no orthogonality constraint imposed). Setting this equal to $\dim S^d(\mathbb{C}^n)$ gives an *expected* generic rank of

$$r_{\text{exp}} := \left\lceil \tfrac{1}{n}\tbinom{n+d-1}{d} \right\rceil.$$

Remark 4.5 (Why this count fails for $d = 2$). Compare Lemma 4.4's count to the matrix warm-up: at $d = 2$, $r_{\text{exp}} = \lceil \binom{p+1}{2}/p \rceil \approx (p+1)/2$, less than the true generic rank p (Corollary 4.3). Given a rank-r symmetric matrix $M = WW^\mathsf{T}$ ($W \in \mathbb{R}^{p\times r}$, $r \leq p$), every other factorization $M = W'(W')^\mathsf{T}$ satisfies $W' = WQ$ for some Q in the orthogonal group $O(r)$ (the same computation as the ICA non-identifiability argument, Proposition 6.2 below): the unconstrained parametrization $(\mathbf{v}_1, \ldots, \mathbf{v}_r) \mapsto \sum \mathbf{v}_i \mathbf{v}_i^\mathsf{T}$ has, for every matrix in its image, a fiber of dimension $\dim O(r) = \binom{r}{2} > 0$. The naive count rn implicitly assumes a *finite* generic fiber, which fails here. For $d \geq 3$, by contrast, the generic fiber is finite (often a single point) for r below a threshold, so the naive count r_{exp} above is accurate for symmetric tensors of order ≥ 3. Generic tensor decompositions, unlike generic matrix decompositions, are unique.

Theorem 4.6 (Alexander–Hirschowitz, 1995)**.** *The symmetric rank of a generic $T \in S^d(\mathbb{C}^n)$ is*

$$\left\lceil \tfrac{1}{n}\tbinom{n+d-1}{d} \right\rceil,$$

except when $d = 2$ (which is generic rank n), or in the cases $(d, n) \in \{(3,5), (4,3), (4,4), (4,5)\}$, where the generic rank is one more than the formula.

Theorem 4.6 was first conjectured in 1901; the case $d = 3$ proved in 1916, the general case by Alexander and Hirschowitz in 1995. See the reference for the complete argument.

Non-Symmetric Tensors

For a general (non-symmetric) CP expression $T = \sum_{i=1}^{r} \mathbf{a}_i^{(1)} \otimes \cdots \otimes \mathbf{a}_i^{(d)}$, $\mathbf{a}_i^{(k)} \in \mathbb{C}^{n_k}$, the analogous count is $r\big(\sum_k n_k - (d-1)\big)$ parameters (each term has n_k free entries per factor, minus $d-1$ scalings), against $\dim \mathbb{C}^{n_1} \otimes \cdots \otimes \mathbb{C}^{n_d} = \prod_k n_k$. The expected generic rank $r_{\text{exp}} = \big\lceil \prod_k n_k / \big(\sum_k n_k - (d-1)\big)\big\rceil$ is not known to hold in general; unlike Theorem 4.6, no analogous exceptional-case classification for non-symmetric tensors is complete.

Remark 4.7 (Open: the smallest unknown case)*.* Theorem 4.6 settles the generic *symmetric* rank for every format. The analogous question for partially symmetric and unbalanced tensors, equivalently, the dimension (or defectivity) of secant varieties of Segre–Veronese varieties, Section 8, is not fully settled; H. Abo's work (with Ottaviani, Peterson, Brambilla, and others) resolves families of cases by induction. They establish that a large family of Segre varieties have the expected secant dimensions for $s \leq 8$, except for a small list. The smallest tensor format not covered by these general results is $2 \times 2 \times 2 \times 3 \times 3$, i.e. $(\mathbb{P}^1)^3 \times (\mathbb{P}^2)^2$. Its expected generic rank is 9. The (open, computational) question is whether $\sigma_9\left((\mathbb{P}^1)^3 \times (\mathbb{P}^2)^2\right)$ fills $\mathbb{P}^{71}$.

5 Moments and Cumulants

Recall from Module 1 that a random vector $\mathbf{z} \in \mathbb{R}^q$ has mean $\mu = \mathbb{E}[\mathbf{z}] \in \mathbb{R}^q$ and covariance $\Sigma_{\mathbf{z}} = \mathbb{E}[(\mathbf{z} - \mu)^{\otimes 2}] \in S^2(\mathbb{R}^q)$. Both are special cases of a general construction.

Definition 5.1 (Moment tensor)**.** The *d-th moment tensor* of $\mathbf{z}$ is $\mathcal{M}_d(\mathbf{z}) := \mathbb{E}[\mathbf{z}^{\otimes d}] \in S^d(\mathbb{R}^q)$, with entries $\mathcal{M}_d(\mathbf{z})_{i_1 \ldots i_d} = \mathbb{E}[z_{i_1} \cdots z_{i_d}]$.

The first moment is the mean: $\mathcal{M}_1(\mathbf{z}) = \mu$; the second moment is the matrix $\mathcal{M}_2(\mathbf{z}) = \mathbb{E}[\mathbf{z}\mathbf{z}^{\mathsf{T}}]$, which relates to the covariance by $\Sigma_{\mathbf{z}} = \mathcal{M}_2(\mathbf{z}) - \mu\mu^{\mathsf{T}}$.

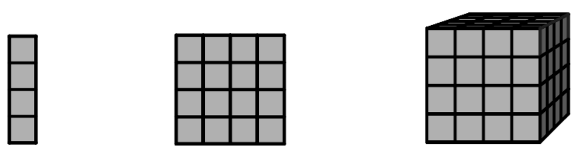

Figure 5: The first moment is a vector, the second is a matrix, the third is a third order tensor.

Definition 5.2. Variables $z_1, \ldots, z_q$ are *independent* if the joint probability density function factorizes as $f_{\mathbf{z}}(t_1, \ldots, t_q) = f_{z_1}(t_1) \cdots f_{z_q}(t_q)$.

Definition 5.3. A tensor T is diagonal if $T_{i_1 \ldots i_d} \neq 0$ implies $i_1 = \cdots = i_d$.

Proposition 5.4. *If $z_1, \ldots, z_q$ are independent with mean zero, then $\mathcal{M}_3(\mathbf{z})$ is diagonal, i.e. $\mathcal{M}_3(\mathbf{z})_{ijk} = 0$ unless $i = j = k$.*

Proof. Independence gives $\mathbb{E}[z_i z_j z_k] = \mathbb{E}[z_i]\mathbb{E}[z_j]\mathbb{E}[z_k]$ whenever i, j, k are not all equal. If i, j, k are pairwise distinct, this is $0 \cdot 0 \cdot 0 = 0$. If two coincide, say $j = k \neq i$, then $\mathbb{E}[z_i z_j^2] = \mathbb{E}[z_i]\mathbb{E}[z_j^2] = 0$ since $\mathbb{E}[z_i] = 0$. □

Remark 5.5 (The same argument fails for $d = 4$)*.* The entry $\mathcal{M}_4(\mathbf{z})_{iijj}$ $(i \neq j)$ equals $\mathbb{E}[z_i^2 z_j^2] = \mathbb{E}[z_i^2]\mathbb{E}[z_j^2]$ by independence, which is generically *non-zero*, since it is a product of two non-negative quantities. So $\mathcal{M}_4(\mathbf{z})$ is not diagonal: independence of the coordinates of $\mathbf{z}$ does not diagonalize higher moment tensors. This is the reason we use *cumulants* instead, see below.

Moment tensors transform under a linear map by a direct generalization of Module 1's congruence action $\Sigma \mapsto A\Sigma A^\mathsf{T}$.

Definition 5.6 (Multiplying a symmetric tensor by a matrix)**.** For $A \in \mathbb{R}^{p\times q}$ and $S \in S^d(\mathbb{R}^q)$, define $A \bullet S \in S^d(\mathbb{R}^p)$ by

$$(A \bullet S)_{i_1\dots i_d} = \sum_{\alpha_1,\dots,\alpha_d=1}^{q} S_{\alpha_1\dots\alpha_d} A_{i_1\alpha_1}\cdots A_{i_d\alpha_d}.$$

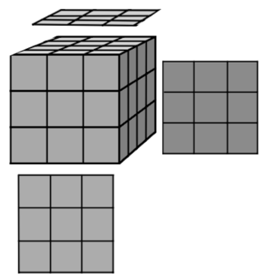

Figure 6: The action $A \bullet S$ multiplies a tensor by a matrix along each index.

At $d = 2$, $(A \bullet S)_{ij} = \sum_{\alpha\beta} S_{\alpha\beta}A_{i\alpha}A_{j\beta} = (ASA^\mathsf{T})_{ij}$: this recovers the congruence action. The expression above is a product on the left and a sum on the right. For a matrix, we saw that a factorization can be viewed as a product or as a sum, too.

Proposition 5.7 (Multilinearity)**.** *Let* $\mathbf{x} = A\mathbf{z}$ *for* $A \in \mathbb{R}^{p\times q}$*. Then* $\mathcal{M}_d(\mathbf{x}) = A \bullet \mathcal{M}_d(\mathbf{z})$*.*

Proof.

$$\mathcal{M}_d(\mathbf{x})_{i_1\dots i_d} = \mathbb{E}[x_{i_1}\cdots x_{i_d}] = \mathbb{E}\Big[\prod_{k=1}^{d}\sum_{\alpha_k} A_{i_k\alpha_k} z_{\alpha_k}\Big] = \sum_{\alpha_1,\dots,\alpha_d} A_{i_1\alpha_1}\cdots A_{i_d\alpha_d}\,\mathbb{E}[z_{\alpha_1}\cdots z_{\alpha_d}],$$

using linearity of expectation. The last factor is $\mathcal{M}_d(\mathbf{z})_{\alpha_1\dots\alpha_d}$, giving $(A \bullet \mathcal{M}_d(\mathbf{z}))_{i_1\dots i_d}$. $\square$

Generating Functions

Definition 5.8 (Moment and cumulant generating functions)**.** For $\mathbf{t} = (t_1,\dots,t_q)^\mathsf{T}$, the *moment generating function* (mgf) of $\mathbf{z}$ is $M_\mathbf{z}(\mathbf{t}) = \mathbb{E}[\exp\langle \mathbf{t},\mathbf{z}\rangle]$, and the *cumulant generating function* (cgf) is $K_\mathbf{z}(\mathbf{t}) = \log M_\mathbf{z}(\mathbf{t})$.

Expanding $\exp\langle \mathbf{t},\mathbf{z}\rangle = \sum_{d\ge 0}\frac{1}{d!}\langle \mathbf{t},\mathbf{z}\rangle^d$ and using $\mathbb{E}\big[\langle \mathbf{t},\mathbf{z}\rangle^d\big] = \langle \mathbf{t}^{\otimes d}, \mathcal{M}_d(\mathbf{z})\rangle$ (expand $\langle \mathbf{t},\mathbf{z}\rangle^d = \sum_{i_1,\dots,i_d} t_{i_1}\cdots t_{i_d} z_{i_1}\cdots z_{i_d}$ and take expectations termwise) gives

$$M_\mathbf{z}(\mathbf{t}) = \sum_{d\ge 0}\frac{1}{d!}\big\langle \mathbf{t}^{\otimes d}, \mathcal{M}_d(\mathbf{z})\big\rangle.$$

Define the *d-th cumulant tensor* $\mathcal{K}_d(\mathbf{z}) \in S^d(\mathbb{R}^q)$ as the analogous coefficient of $K_\mathbf{z}$:

$$K_\mathbf{z}(\mathbf{t}) = \sum_{d\ge 1}\frac{1}{d!}\big\langle \mathbf{t}^{\otimes d}, \mathcal{K}_d(\mathbf{z})\big\rangle.$$

Proposition 5.9. *The first two cumulants are $\mathcal{K}_1(\mathbf{z}) = \mu$ and $\mathcal{K}_2(\mathbf{z}) = \Sigma_{\mathbf{z}}$.*

Proof. Write $K_{\mathbf{z}}(\mathbf{t}) = \log\big(1+u\big)$ with $u = \langle \mathbf{t}, \mu\rangle + \frac{1}{2}\langle \mathbf{t}^{\otimes 2}, \mathcal{M}_2(\mathbf{z})\rangle + O(\mathbf{t}^3)$, and expand $\log(1+u) = u - \frac{1}{2}u^2 + O(u^3)$. The degree-1 part of $\log(1+u)$ is $\langle \mathbf{t}, \mu\rangle$, giving $\mathcal{K}_1(\mathbf{z}) = \mu$. The degree-2 part is $\frac{1}{2}\langle \mathbf{t}^{\otimes 2}, \mathcal{M}_2(\mathbf{z})\rangle - \frac{1}{2}\langle \mathbf{t}, \mu\rangle^2 = \frac{1}{2}\big\langle \mathbf{t}^{\otimes 2}, \mathcal{M}_2(\mathbf{z}) - \mu^{\otimes 2}\big\rangle$ (using $\langle \mathbf{t}, \mu\rangle^2 = \langle \mathbf{t}^{\otimes 2}, \mu^{\otimes 2}\rangle$), so $\mathcal{K}_2(\mathbf{z}) = \mathcal{M}_2(\mathbf{z}) - \mu^{\otimes 2} = \mathbb{E}[(\mathbf{z}-\mu)^{\otimes 2}] = \Sigma_{\mathbf{z}}$. ◻

Proposition 5.10 (The cumulant tensors of independent variables are diagonal)**.** *If $z_1, \ldots, z_q$ are independent, $\mathcal{K}_d(\mathbf{z})$ is diagonal for every d.*

Proof. Independence gives $M_{\mathbf{z}}(\mathbf{t}) = \mathbb{E}\big[\exp(t_1 z_1)\cdots\exp(t_q z_q)\big] = \prod_{i=1}^q \mathbb{E}[\exp(t_i z_i)]$, so

$$K_{\mathbf{z}}(\mathbf{t}) = \log M_{\mathbf{z}}(\mathbf{t}) = \sum_{i=1}^{q} \log \mathbb{E}[\exp(t_i z_i)],$$

a sum of functions each depending on a *single* coordinate t_i. Consequently no monomial $t_i t_j \cdots$ with two or more distinct indices can appear in the Taylor expansion of $K_{\mathbf{z}}$ at any degree, i.e. the degree-d coefficient tensor $\mathcal{K}_d(\mathbf{z})$ has entries supported only on indices $(i, i, \ldots, i)$. ◻

Proposition 5.10 is the resolution of Remark 5.5: cumulants, unlike moments, diagonalize under independence at *every* order, not just order 3. Passing from moments to cumulants can be thought of as the higher-order analog of mean-centering.

6 Independent Component Analysis

Definition 6.1 (Independent Component Analysis)**.** *Independent Component Analysis* (ICA) models an observed random vector $\mathbf{x} = (x_1, \ldots, x_p)^\top$ as $\mathbf{x} = A\mathbf{z}$ for a fixed, unknown, invertible mixing matrix $A \in \mathbb{R}^{p\times p}$ and a latent random vector $\mathbf{z} = (z_1, \ldots, z_p)^\top$ with *independent* entries. (We take $p = q$, the classical case; Section 8 discusses $q \neq p$.)

This is the latent-variable model $\mathbf{x} = A\mathbf{z}$, with the entries of $\mathbf{z}$ independent rather than just uncorrelated. The classical motivating picture is the *cocktail party problem*: p microphones each record a mixture $x_i = \sum_j A_{ij} z_j$ of p underlying, independent sound sources z_j; the goal is to recover A and the sources from the microphone recordings alone (blind source separation).

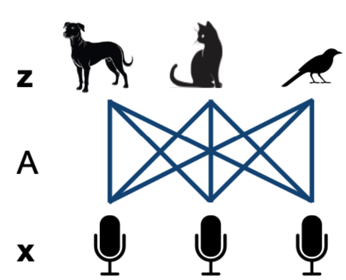

Figure 7: ICA for the cocktail party problem. Sources $\mathbf{z}$ are mixed together via mixing matrix A to produce observed variables $\mathbf{x}$.

Independence implies that all cumulant tensors are diagonal. If we only impose that the second cumulant (covariance) is diagonal, we recover the uncorrelated assumption beyind PCA. We saw in the first part of the course that this yields a matrix factorization problem $\Sigma_{\mathbf{x}} = A\Sigma_{\mathbf{z}}A^{\mathsf{T}}$, which is solved by the spectral theorem for orthogonal A. If we don't impose orthogonality of A, then we will see below that we only identify A up to an orthogonal transformation: any $A' = AQ$, for Q orthogonal, gives a model with equally good fit.

Why the Covariance Alone Is Not Enough

Since the z_j are independent, $\Sigma_{\mathbf{z}}$ is diagonal, so $\Sigma_{\mathbf{x}} = A\Sigma_{\mathbf{z}}A^{\mathsf{T}} = ADA^{\mathsf{T}}$ for some diagonal $D \succeq 0$. Given (an estimate of) $\Sigma_{\mathbf{x}}$, can we recover A and D?

Proposition 6.2. *Let $\Sigma \in S^2(\mathbb{R}^p)$, $\Sigma \succeq 0$. The solution set $\{(A, D) : A \in \mathbb{R}^{p\times p}$ invertible, $D \succeq 0$ diagonal, $\Sigma = ADA^{\mathsf{T}}\}$, if non-empty, is*

$$\{(B_0 Q\Lambda, \Lambda^{-2}) : Q \in O(p),\ \Lambda \text{ diagonal invertible}\}$$

for any one solution $B_0 = A_0 D_0^{1/2}$.

Proof. Write $B = AD^{1/2}$, so $\Sigma = BB^{\mathsf{T}}$. If $B_0 B_0^{\mathsf{T}} = BB^{\mathsf{T}} = \Sigma$, then $(B_0^{-1}B)(B_0^{-1}B)^{\mathsf{T}} = B_0^{-1}\Sigma(B_0^{\mathsf{T}})^{-1} = B_0^{-1}(B_0B_0^{\mathsf{T}})(B_0^{\mathsf{T}})^{-1} = I$, so $Q := B_0^{-1}B$ is orthogonal and $B = B_0Q$. Since $B = AD^{1/2}$, any positive rescaling of B's columns can be absorbed into $D^{1/2}$; writing this rescaling as Λ gives $(A, D) = (B_0Q\Lambda, \Lambda^{-2})$. □

The solution set has dimension $\dim O(p) + p = \binom{p}{2} + p$ (an arbitrary rotation, plus rescaling each of p columns), strictly positive for $p \geq 2$: $\Sigma_{\mathbf{x}}$ alone never pins down (A, D). As a consistency check: the naive parametrization (A, D) has $p^2 + p$ free entries, mapping into $\Sigma \in S^2(\mathbb{R}^p)$, of dimension $\binom{p+1}{2}$; a generic fiber of a map between spaces of these dimensions is expected to have dimension $(p^2 + p) - \binom{p+1}{2} = \binom{p}{2} + p$, matching the solution set above.

Diagonalizing a Higher Cumulant Instead

Recovering A requires looking beyond $\Sigma_{\mathbf{x}}$. Perhaps, then, we can ammend Benzécri's statement to say that "all data analysis is tensor diagonalization; the crux of the matter is in finding the right tensor to diagonalize." Having diagonalized the covariance, we now diagonalize a higher cumulant.

Proposition 6.3. *Fix $d \geq 3$. Writing $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_p]$ (columns) and $\lambda_i := \mathcal{K}_d(\mathbf{z})_{i,i,\ldots,i}$ (d copies of i; the i-th source's own order-d cumulant),*

$$\mathcal{K}_d(\mathbf{x}) = A \bullet \mathcal{K}_d(\mathbf{z}) = \sum_{i=1}^{p} \lambda_i\, \mathbf{a}_i^{\otimes d}.$$

Proof. By Proposition 5.10, $\mathcal{K}_d(\mathbf{z}) = \sum_{i=1}^p \lambda_i\, \mathbf{e}_i^{\otimes d}$, diagonal. By the cumulant analog of multilinearity of moments (Proposition 5.7; the identical proof applies to cgf Taylor coefficients), $\mathcal{K}_d(\mathbf{x}) = A \bullet \mathcal{K}_d(\mathbf{z}) = \sum_i \lambda_i\, (A \bullet \mathbf{e}_i^{\otimes d})$. Directly from the definition of $\bullet$, $(A \bullet \mathbf{e}_i^{\otimes d})_{j_1 \ldots j_d} = A_{j_1 i} \cdots A_{j_d i} = (\mathbf{a}_i^{\otimes d})_{j_1 \ldots j_d}$, so $A \bullet \mathbf{e}_i^{\otimes d} = \mathbf{a}_i^{\otimes d}$. □

Proposition 6.3 is the point of this module: *ICA is a symmetric tensor decomposition problem.* Recovering A from $\mathbf{x}$'s d-th cumulant means recovering the rank-one terms of a symmetric CP decomposition, and the columns of A are its factors. Sections 7 and 8 ask when the decomposition is unique.

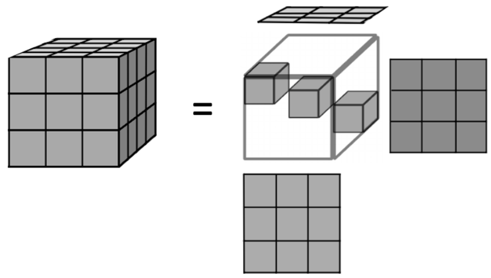

Figure 8: The change of basis from Figure 7 specializes to give a sum of outer products of vectors as in Figure 1.

7 Identifiability of ICA

The best we can hope to recover is A up to permuting and rescaling its columns: if $\mathbf{x} = A\mathbf{z}$, then for any permutation matrix P and invertible diagonal Λ, $\mathbf{x} = (AP\Lambda)(\Lambda^{-1}P^\mathsf{T}\mathbf{z})$, and $\Lambda^{-1}P^\mathsf{T}\mathbf{z}$ is again a vector of independent variables. So "identifiable" means: recoverable up to this ambiguity.

Definition 7.1. A random variable z is *Gaussian* with mean μ, variance σ^2 if it has density $f(x) = (2\pi\sigma^2)^{-1/2}\exp\big(-\frac{1}{2\sigma^2}(x-\mu)^2\big)$; write $z \sim N(\mu, \sigma^2)$.

Theorem 7.2 (Marcinkiewicz, 1939)**.** *A Gaussian variable $z \sim N(\mu, \sigma^2)$ has cgf $K_z(t) = \mu t + \frac{1}{2}\sigma^2 t^2$. Conversely, if a random variable has a cumulant generating function that is finite for all t and is a polynomial, then it is Gaussian.*

A Gaussian variable has $\kappa_d(z) = 0$ for every $d \geq 3$. By Proposition 6.3, if every source z_i is Gaussian, $\mathcal{K}_d(\mathbf{x}) = 0$ for every $d \geq 3$: Gaussian sources are invisible to the higher-order statistics.

Suppose $\mathbf{x} = A\mathbf{z} = A'\mathbf{z}'$ for two invertible mixing matrices and two vectors of independent sources. Then $\mathbf{z} = (A^{-1}A')\mathbf{z}' =: B\mathbf{z}'$, so B is an invertible matrix sending one vector of independent variables, $\mathbf{z}'$, to another, $\mathbf{z}$.

Theorem 7.3 (Darmois–Skitovich, 1953)**.** *Let $\mathbf{z} = (z_1, \ldots, z_p)$ be independent, and let $\sum_i a_i z_i$ and $\sum_i b_i z_i$ be independent (for fixed scalars a_i, b_i). Then every z_i with $a_i b_i \neq 0$ is Gaussian.*

Proposition 7.4. *Let $\mathbf{z}' = (z'_1, \ldots, z'_p)$ be independent and $B \in \mathbb{R}^{p\times p}$ invertible, with $B\mathbf{z}'$ also independent. Then for every source index i, either z'_i is Gaussian, or column i of B has exactly one non-zero entry.*

Proof. For any two rows $j \neq k$, $(B\mathbf{z}')_j = \sum_i B_{ji}z'_i$ and $(B\mathbf{z}')_k = \sum_i B_{ki}z'_i$ are independent (two coordinates of an independent vector). By Theorem 7.3, for every i: either $B_{ji}B_{ki} = 0$, or z'_i is Gaussian. If z'_i is not Gaussian, this holds for *every* pair $j \neq k$, which forces at most one row j with $B_{ji} \neq 0$ (two non-zero rows j, k in column i would violate $B_{ji}B_{ki} = 0$). Since B is invertible, column i is non-zero, so it has exactly one non-zero entry. □

Theorem 7.5 (Comon, 1994)**.** *Fix $\mathbf{x} = A\mathbf{z}$, A invertible, with $\mathbf{z} = (z_1, \ldots, z_p)$ independent. The ICA model is identifiable (i.e. A is recoverable up to permutation and scaling of columns) if and only if at most one z_i is Gaussian.*

We prove Theorem 7.5 in three parts, following the setup above: $B = A^{-1}A'$ is invertible and sends the independent vector $\mathbf{z}'$ to the independent vector $\mathbf{z} = B\mathbf{z}'$.

Part 1: no Gaussian sources ⇒ identifiable. By Proposition 7.4, every column of B has one non-zero entry, say column i's non-zero entry is in row $r(i)$. If $r(i) = r(i')$ for $i \neq i'$, columns i, i' are both scalar multiples of $\mathbf{e}_{r(i)}$, hence linearly dependent, contradicting invertibility of B; so r is injective, hence (as a self-map of a finite set) a bijection. So $B = P\Lambda$ for a permutation matrix P and invertible diagonal Λ: the trivial ambiguity. □

Part 2: two or more Gaussian sources ⇒ not identifiable. Suppose z_1', z_2' are both Gaussian and independent; rescale so both are $N(0,1)$. The linear map $B_0 = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ (identity on all other coordinates) sends (z_1', z_2') to $(z_1' + z_2', z_1' - z_2')$, jointly Gaussian (a linear map of a Gaussian vector) with covariance $B_0 B_0^{\mathsf{T}} = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 2 \end{smallmatrix}\right)$. A jointly Gaussian vector with diagonal covariance has independent entries: its density $f(\mathbf{x}) \propto \exp(-\frac{1}{2}\mathbf{x}^{\mathsf{T}}\Sigma^{-1}\mathbf{x})$ factors into a product of one-dimensional Gaussian densities when Σ (equivalently Σ^{-1}) is diagonal. So $B_0\mathbf{z}'$ is again a vector of independent variables, but B_0 is not a signed permutation: this is a different, valid solution, so A is not identifiable. (If $p > 2$, extend B_0 by the identity on the remaining coordinates; the argument is unchanged.) □

Part 3: one Gaussian source ⇒ identifiable. Reorder the variables so z_p' is the (unique) Gaussian source; $z_1', \ldots, z_{p-1}'$ are not Gaussian. By Proposition 7.4, columns $1, \ldots, p-1$ of B each have one non-zero entry, and (as in Part 1) these must lie in $p-1$ distinct rows; permute rows so column i's non-zero entry is in row i, for $i = 1, \ldots, p-1$, and rescale rows so each such entry equals 1. Since B is invertible and columns $1, \ldots, p-1$ already occupy rows $1, \ldots, p-1$ (each with a single entry), row p must be zero in columns $1, \ldots, p-1$, and column p has an entry $\lambda_p \neq 0$ in row p (else B singular). Write column p's other entries as $\lambda_1, \ldots, \lambda_{p-1}$, so

$$(B\mathbf{z}')_i = z_i' + \lambda_i z_p' \quad (i < p), \qquad (B\mathbf{z}')_p = \lambda_p z_p'.$$

Since $B\mathbf{z}'$ is independent, $(B\mathbf{z}')_i \perp (B\mathbf{z}')_p$ for each $i < p$, in particular uncorrelated:

$$0 = \operatorname{Cov}\left(z_i' + \lambda_i z_p',\ \lambda_p z_p'\right) = \lambda_p\left[\operatorname{Cov}(z_i', z_p') + \lambda_i \operatorname{Var}(z_p')\right] = \lambda_p \lambda_i \operatorname{Var}(z_p'),$$

using $z_i' \perp z_p'$. Since $\lambda_p \neq 0$ and $\operatorname{Var}(z_p') > 0$ (non-degenerate), $\lambda_i = 0$ for every $i < p$. So B (after undoing the row permutation) is diagonal, hence a signed permutation. □

8 Existence and Uniqueness of Tensor Decompositions

In practice we have only a data matrix $X \in \mathbb{R}^{n \times p}$, and must estimate $\mathcal{K}_d(\mathbf{x})$ from samples before decomposing it. This raises a different question: does the tensor $\mathcal{K}_d(\mathbf{x}) = \sum_{i=1}^{p} \lambda_i \mathbf{a}_i^{\otimes d}$, viewed simply as a symmetric tensor in $S^d(\mathbb{R}^p)$ of rank p, have a *unique* such decomposition?

The number of equations available (matching entries of $\mathcal{K}_d(\mathbf{x})$) is $\dim S^d(\mathbb{R}^p) = \binom{p+d-1}{d}$; the number of unknowns in A (with $q \leq p$ sources) is qp. Larger d gives more equations relative to unknowns, at the cost of a d-th cumulant that is harder to estimate accurately from finite samples. This is a practical tradeoff in choosing d.

Theorem 8.1 (Harshman, 1970)**.** *Let $T = \sum_{i=1}^{r} \lambda_i \mathbf{a}_i^{\otimes d} \in S^d(\mathbb{R}^p)$, $A = [\mathbf{a}_1 \cdots \mathbf{a}_r]$. If $r = \operatorname{rank}(A)$ (in particular $r \leq p$), this decomposition is the unique one, up to permutation and rescaling of terms.*

Exercise 8.2. Let $T = \mathbf{e}_1^{\otimes 3} + \mathbf{e}_2^{\otimes 3} \in S^3(\mathbb{R}^p)$, $p \geq 2$. Identify the factor matrix A to which Theorem 8.1 applies, and check $\operatorname{rank}(A) = 2$.

Theorem 8.1's hypothesis $r = \operatorname{rank}(A)$ is restrictive: it requires A's columns to be linearly independent, hence $r \leq p$. A weaker, more robust condition suffices.

Definition 8.3 (Kruskal rank)**.** The *Kruskal rank* of $A \in \mathbb{R}^{p\times r}$, denoted k_A, is the largest integer k such that every k columns of A are linearly independent.

Always $k_A \leq \operatorname{rank}(A) \leq p$; unlike ordinary rank, k_A can be strictly less than $\operatorname{rank}(A)$ even for a full-rank A with $r > p$ columns (e.g. four columns in $\mathbb{R}^3$, no three of which are dependent, but some four necessarily are).

Theorem 8.4 (Kruskal, 1977)**.** *If $r \leq \frac{d}{2}k_A - 1$, the decomposition $T = \sum_{i=1}^{r} \lambda_i \mathbf{a}_i^{\otimes d}$ is the unique one, up to permutation and rescaling of terms.*

Generic Uniqueness

Theorems 8.1 and 8.4 certify uniqueness for a *specific* tensor, given information about its factor matrix. A different, and in this module's context more striking, question is whether uniqueness holds for *almost every* tensor of a given rank, with no assumption on the factors at all.

Theorem 8.5 (Chiantini–Ottaviani–Vannieuwenhoven, 2017)**.** *A generic $T \in S^d(\mathbb{C}^n)$ of rank strictly less than the Alexander–Hirschowitz threshold $\frac{1}{n}\binom{n+d-1}{d}$ (Theorem 4.6) has a* unique *symmetric rank decomposition, except for three specific formats, $S^6(\mathbb{C}^3)$ at rank 9, $S^4(\mathbb{C}^4)$ at rank 8, $S^3(\mathbb{C}^6)$ at rank 9, each of which has two decompositions.*

Together with Remark 4.5, this is the module's central identifiability contrast with Module 1: a *generic* low-rank symmetric tensor of order $d \geq 3$ is already uniquely decomposable, with no extra structural assumption (such as orthogonality) required, unlike a symmetric matrix, whose generic decomposition always has a positive-dimensional family of alternatives (Remark 4.5).

The Algebraic Geometry of Secant Varieties

Theorem 8.5 is a statement in classical algebraic geometry. We record the minimal vocabulary needed to state it precisely.

Definition 8.6. A *variety* is a subset of $\mathbb{C}^N$ defined by the vanishing of a collection of polynomial equations. A *limit point* of a set X is a point of the form $\lim_{\varepsilon\to 0}\mathbf{x}_\varepsilon$ for a convergent sequence $\mathbf{x}_\varepsilon \in X$; the *closure* $\overline{X}$ of X is X together with all its limit points.

Definition 8.7. The *Veronese variety* $\nu_d(\mathbb{C}^n) \subseteq S^d(\mathbb{C}^n)$ is the set of symmetric tensors of rank ≤ 1. The *Segre variety* $\operatorname{Seg}(\mathbb{C}^{n_1} \times \cdots \times \mathbb{C}^{n_d}) \subseteq \mathbb{C}^{n_1} \otimes \cdots \otimes \mathbb{C}^{n_d}$ is the set of (not necessarily symmetric) tensors of rank ≤ 1.

Definition 8.8. The *r-th secant variety* of the Veronese, $\sigma_r(\nu_d(\mathbb{C}^n)) \subseteq S^d(\mathbb{C}^n)$, is the closure of the set of symmetric tensors of rank $\leq r$. Likewise, $\sigma_r(\operatorname{Seg}(\mathbb{C}^{n_1} \times \cdots \times \mathbb{C}^{n_d}))$ is the closure of the set of tensors of rank $\leq r$.

Why is the closure needed: why can a tensor's rank jump under a limit? The standard example is: $\mathbf{v} = \mathbf{e}_1, \mathbf{w} = \mathbf{e}_2 \in \mathbb{C}^2$,

$$\lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\Big[(\mathbf{v}+\varepsilon\mathbf{w})^{\otimes 3} - \mathbf{v}^{\otimes 3}\Big] = \mathbf{v}\otimes\mathbf{v}\otimes\mathbf{w} + \mathbf{v}\otimes\mathbf{w}\otimes\mathbf{v} + \mathbf{w}\otimes\mathbf{v}\otimes\mathbf{v},$$

by expanding $(\mathbf{v}+\varepsilon\mathbf{w})^{\otimes 3} = \mathbf{v}^{\otimes 3} + \varepsilon(\mathbf{v}\otimes\mathbf{v}\otimes\mathbf{w} + \mathbf{v}\otimes\mathbf{w}\otimes\mathbf{v} + \mathbf{w}\otimes\mathbf{v}\otimes\mathbf{v}) + O(\varepsilon^2)$. For each $\varepsilon \neq 0$, the left-hand bracket, before dividing by ε, is a difference of two rank-one tensors, hence of rank ≤ 2; the limiting tensor $T = \mathbf{v}\otimes\mathbf{v}\otimes\mathbf{w} + \mathbf{v}\otimes\mathbf{w}\otimes\mathbf{v} + \mathbf{w}\otimes\mathbf{v}\otimes\mathbf{v}$ therefore lies in $\sigma_2(\operatorname{Seg}(\mathbb{C}^2\times\mathbb{C}^2\times\mathbb{C}^2))$, yet $\operatorname{rank}(T) = 3$: T itself is not a sum of 2 rank-one tensors, only a limit of such sums.

9 Optimization Landscapes for Tensors

We have so far looked at the existence (Theorem 4.6) and uniqueness (Theorems 8.1–8.5) of a tensor decomposition. This section gets closer to computing a decomposition. Recall from Module 1 that the top principal component is the maximizer of $\mathbf{u}^\mathsf{T} S\mathbf{u}$ over $\|\mathbf{u}\| = 1$, and that its Lagrangian critical point condition, $S\mathbf{u} = \lambda\mathbf{u}$, is the eigenvector equation. We generalize this to symmetric tensors.

Proposition 9.1. *Let $T \in S^3(\mathbb{R}^n)$ and consider $\max_{\|\mathbf{v}\|=1} T(\mathbf{v},\mathbf{v},\mathbf{v})$, where $T(\mathbf{v},\mathbf{v},\mathbf{v}) = \sum_{ijk} T_{ijk}v_iv_jv_k$. The critical points of the Lagrangian are the $\mathbf{v}$ satisfying (for some scalar μ)*

$$T(\cdot,\mathbf{v},\mathbf{v}) \parallel \mathbf{v}, \qquad \textit{i.e.} \qquad \sum_{jk} T_{ljk}v_jv_k = \mu\, v_l \quad \forall l.$$

Proof. The Lagrangian is $L(\mathbf{v},\mu) = \sum_{ijk} T_{ijk}v_iv_jv_k - \mu(\|\mathbf{v}\|^2 - 1)$. Differentiating the cubic form by the product rule,

$$\frac{\partial}{\partial v_l}\sum_{ijk} T_{ijk}v_iv_jv_k = \sum_{jk} T_{ljk}v_jv_k + \sum_{ik} T_{ilk}v_iv_k + \sum_{ij} T_{ijl}v_iv_j = 3\sum_{jk} T_{ljk}v_jv_k,$$

where the three sums are equal by symmetry of T (relabel dummy indices in the second and third sums). So $\partial L/\partial v_l = 3\sum_{jk} T_{ljk}v_jv_k - 2\mu v_l = 0$ for all l, i.e. $T(\cdot,\mathbf{v},\mathbf{v}) = \frac{2\mu}{3}\mathbf{v}$: proportional to $\mathbf{v}$. □

Definition 9.2 (Eigenvector of a tensor)**.** A unit $\mathbf{v}$ with $T(\cdot,\mathbf{v},\mathbf{v}) = \mu\mathbf{v}$ for some scalar μ is an *eigenvector* of $T \in S^3(\mathbb{R}^n)$, with eigenvalue μ. (The same definition, with the identical proof, applies to $T \in S^d(\mathbb{R}^n)$ for any $d \geq 2$; at $d = 2$ it recovers the matrix eigenvector equation $S\mathbf{v} = \mu\mathbf{v}$.)

Proposition 9.3 (Eigenvectors give the best rank-one approximation)**.** *The unit vector $\mathbf{v}$ maximizing $T(\mathbf{v},\mathbf{v},\mathbf{v})^2$ also minimizes $\|T - \lambda\mathbf{v}^{\otimes 3}\|$ over $\|\mathbf{v}\| = 1$, $\lambda \in \mathbb{R}$; both are eigenvectors of T.*

Proof. For unit $\mathbf{v}$, $\|T - \lambda\mathbf{v}^{\otimes 3}\|^2 = \langle T, T\rangle - 2\lambda\langle T, \mathbf{v}^{\otimes 3}\rangle + \lambda^2\langle \mathbf{v}^{\otimes 3}, \mathbf{v}^{\otimes 3}\rangle = \|T\|^2 - 2\lambda\, T(\mathbf{v},\mathbf{v},\mathbf{v}) + \lambda^2$, using $\|\mathbf{v}\| = 1$. Minimizing over λ: $\partial_\lambda(\cdots) = -2T(\mathbf{v},\mathbf{v},\mathbf{v}) + 2\lambda = 0 \Rightarrow \lambda = T(\mathbf{v},\mathbf{v},\mathbf{v})$, and substituting back leaves $-T(\mathbf{v},\mathbf{v},\mathbf{v})^2$ to minimize, i.e. $T(\mathbf{v},\mathbf{v},\mathbf{v})^2$ to maximize. Since $T(\mathbf{v},\mathbf{v},\mathbf{v})^2$ is a smooth non-negative function of $\mathbf{v}$ whose critical points coincide with those of $T(\mathbf{v},\mathbf{v},\mathbf{v})$ itself (by the chain rule, away from $T(\mathbf{v},\mathbf{v},\mathbf{v}) = 0$), its maximizer is an eigenvector of T by Proposition 9.1. □

Remark 9.4. In the matrix case ($d = 2$), we have $(-\mathbf{v})^\mathsf{T} S(-\mathbf{v}) = \mathbf{v}^\mathsf{T} S\mathbf{v}$. A PSD matrix achieves $\mathbf{v}^\mathsf{T} S\mathbf{v} \geq 0$ for every $\mathbf{v}$ (Module 1). Similar conditions can be imposed for even order tensors. However, for odd tensors, sign-definiteness cannot be imposed. For example, when $d = 3$ we have $T(-\mathbf{v},-\mathbf{v},-\mathbf{v}) = -T(\mathbf{v},\mathbf{v},\mathbf{v})$. A non-zero $T \in S^3(\mathbb{R}^n)$ cannot satisfy $T(\mathbf{v},\mathbf{v},\mathbf{v}) \geq 0$ for all $\mathbf{v}$. This is why Proposition 9.3 maximizes $T(\mathbf{v},\mathbf{v},\mathbf{v})^2$ rather than $T(\mathbf{v},\mathbf{v},\mathbf{v})$.

Remark 9.5 (Singular vectors of general tensors). The same variational idea, without symmetry, defines *singular vectors* of a general $T \in \mathbb{R}^{n_1} \otimes \mathbb{R}^{n_2} \otimes \mathbb{R}^{n_3}$: unit vectors $(\mathbf{u},\mathbf{v},\mathbf{w})$ that are critical points of $T(\mathbf{u},\mathbf{v},\mathbf{w})$ subject to $\|\mathbf{u}\| = \|\mathbf{v}\| = \|\mathbf{w}\| = 1$, directly generalizing the singular vector pairs of Module 1's SVD. We do not derive this further here.

10 Outlook: Beyond Tensor Decomposition

Module 1 closed by asking what other matrices are worth diagonalizing, and named the fourth cumulant, a $p \times p \times p \times p$ array, as a first example. This module has extended this example to other tensors and their decompositions. We have studied the multi-linear algebra of tensors, developed from three definitions of a tensor, gives the vocabulary needed to diagonalize them, and studied ICA as the resulting model. Its identifiability rests on a new phenomenon, absent from the matrix case: a generic low-rank tensor of order ≥ 3 is already uniquely decomposable (Theorem 8.5), where a generic symmetric matrix never is (Remark 4.5).

This is not specific to ICA. Spectral and moment-based methods for estimating latent-variable and mixture models use it: build a higher moment or cumulant tensor of the observed data, and decompose it, using Kruskal- or genericity-type uniqueness (Theorems 8.4, 8.5) to recover parameters that pairwise, covariance-level statistics cannot.

References

- J. M. Landsberg, *Tensors: Geometry and Applications*, American Mathematical Society, 2012.
- T. G. Kolda & B. W. Bader, "Tensor Decompositions and Applications," *SIAM Review* 51(3), 2009.
- A. Hyvärinen, J. Karhunen, & E. Oja, *Independent Component Analysis*, Wiley, 2001.
- L.-H. Lim, "Tensors in computations," *Acta Numerica* 30, 2021 (arXiv:2106.08090).
- A. Seigal, "Factorizations for Data Analysis," *SIAM News*, Vol. 58, No. 7, September 2025.
- J. J. Sylvester, "On the principles of the calculus of forms," 1851.
- J. Marcinkiewicz, "Sur une propriété de la loi de Gauss," *Math. Zeitschrift* 44, 1939.
- M. Kac, "On a characterization of the normal distribution," *Amer. J. Math.* 61, 1939; S. Bernstein, 1941.
- G. Darmois, "Analyse générale des liaisons stochastiques," *Rev. Inst. Internat. Stat.* 21, 1953; V. P. Skitovich, 1953.
- P. Comon, "Independent component analysis, a new concept?" *Signal Processing* 36(3), 1994.
- R. A. Harshman, "Foundations of the PARAFAC procedure," *UCLA Working Papers in Phonetics* 16, 1970.
- J. B. Kruskal, "Three-way arrays: rank and uniqueness of trilinear decompositions," *Linear Algebra Appl.* 18, 1977.
- J. Alexander & A. Hirschowitz, "Polynomial interpolation in several variables," *J. Algebraic Geom.* 4, 1995.
- L. Chiantini, G. Ottaviani, & N. Vannieuwenhoven, "Effective criteria for specific identifiability of tensors and forms," *SIAM J. Matrix Anal. Appl.* 38(2), 2017.
- A. Fawzi et al., "Discovering faster matrix multiplication algorithms with reinforcement learning," *Nature* 610, 2022.

- Google DeepMind, "AlphaEvolve: A coding agent for scientific and algorithmic discovery," 2025.